

Altermagnetism in RuO_2

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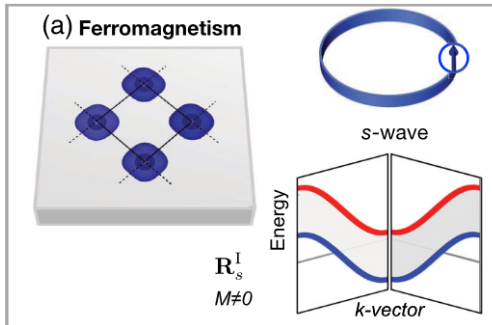
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What is Altermagnetism?

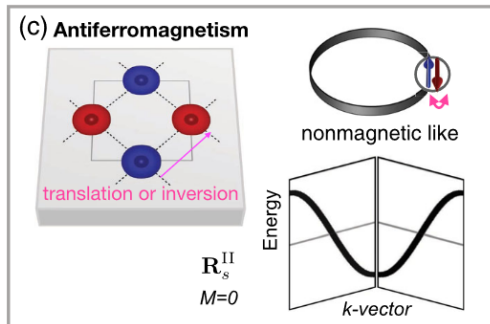
Altermagnetism: The 3rd Magnetic state



- Ferromagnetic: Spin-split bands
 - Antiferromagnetic: Anti-aligned structure → No net magnetization!
 - Altermagnetic: Anti aligned structure + spin splitting!¹
- New state! → Applications in spintronics

¹Šmejkal et al., Physics Review X, 2022

Altermagnetism: The 3rd Magnetic state

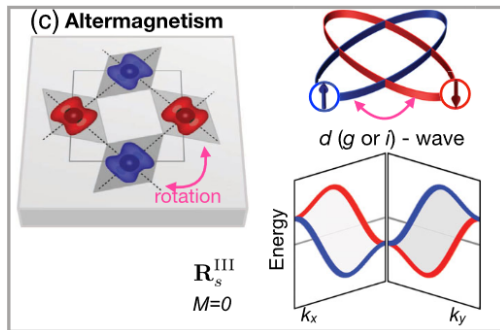


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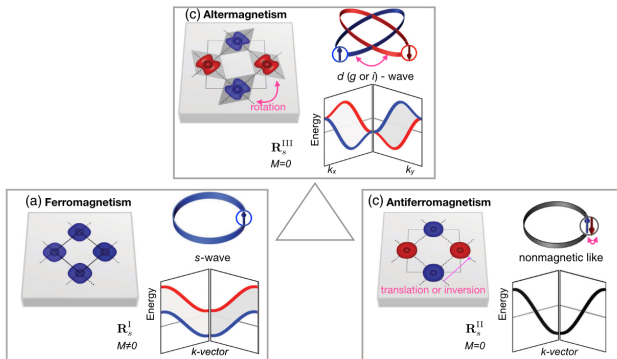


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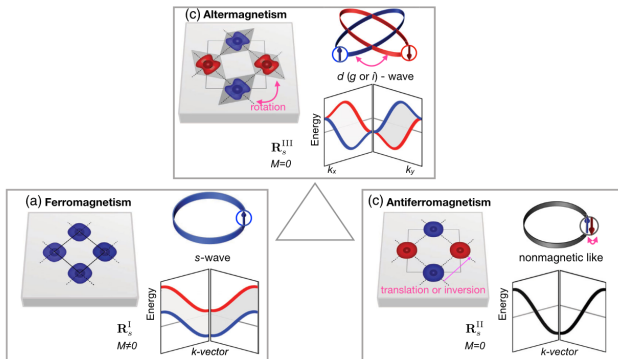


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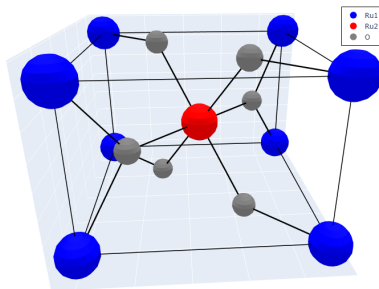


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Altermagnetism in RuO₂



Altermagnetism has been confirmed:

- Theoretically
- Experimentally

Motivation

Goals

It is my turn:

- Confirm altermagnetism in RuO_2 by density functional theory

Understand:

- The novel properties
- The topological properties

To understand it:

→ Define and evaluate calculate the Berry curvature!

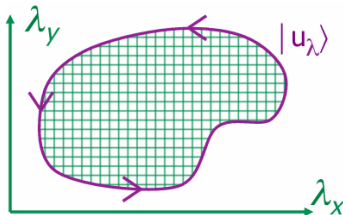
Berry Curvature

Berry connection in terms of Bloch function²:

$$\mathbf{A}_n(\mathbf{k}) = i \langle u_{n\mathbf{k}} | \nabla_{\mathbf{k}} | u_{n\mathbf{k}} \rangle ,$$

Berry curvature:

$$\begin{aligned}\Omega_n(\mathbf{k}) &= \nabla_{\mathbf{k}} \times \mathbf{A}_n(\mathbf{k}) \\ &= -\text{Im} \langle \nabla_{\mathbf{k}} u_{n\mathbf{k}} | \times | u_{n\mathbf{k}} \rangle\end{aligned}$$



Or in Kubo-like expression:

$$\Omega_\gamma(\mathbf{k}) = -2\varepsilon^{\alpha\beta\gamma} \text{Im} \sum_{m \neq n}^{n_{\text{occ.}}} \frac{\langle u_{n\mathbf{k}} | \hat{v}_\alpha | u_{m\mathbf{k}} \rangle \langle u_{m\mathbf{k}} | \hat{v}_\beta | u_{n\mathbf{k}} \rangle}{(E_{m\mathbf{k}} - E_{n\mathbf{k}})^2}$$

Berry curvature can come from:

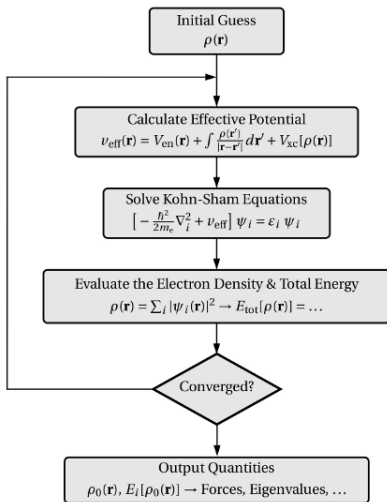
- Symmetry of crystal
- Spin-orbit coupling (SOC) ← Commonly been used

What if we only consider the effect of the symmetry of crystal (no SOC)?

²<http://www-dft.ts.infn.it/resta/gtse/draft.pdf>

Methods

Density Functional Theory



Exchange functional:

- General gradient approximation

$$E_{XC}^{GGA}[n_i] = \int \varepsilon_{XC}(n_i, \nabla n_i) n(\mathbf{r}) d^3\mathbf{r}$$

Empirical parameter:

- Ru d-orbital electron: on-site spin repulsion.
- Focus in spin-polarization.

Hubbard parameter U.

Therefore,

→ DFT + U formalism

Wannier function - Wannier Basis

Isolated Wannier function from Bloch wavefunction:

$$\Phi_n(\mathbf{r} - \mathbf{R}) = \Phi_{n\mathbf{R}}(\mathbf{r}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} u_{\mathbf{k}}(\mathbf{r}) e^{i\mathbf{k}(\mathbf{r}-\mathbf{R})}$$

Non uniqueness of Wannier under transformation:

$$u_{\mathbf{k}}(\mathbf{r}) \rightarrow e^{i\varphi_{\mathbf{k}}} u_{\mathbf{k}}(\mathbf{r}) \quad (1)$$

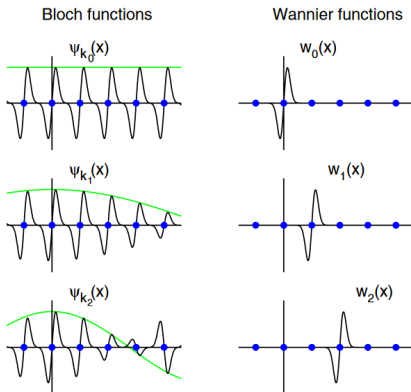
Spread of the function:

$$\sigma^2 = \sum_n \left(\langle r^2 \rangle_n - \langle r \rangle_n^2 \right) = \sigma_I^2 + \tilde{\sigma}^2,$$

Under (1):

- σ_I^2 invariant
- $\tilde{\sigma}^2$ increase spread

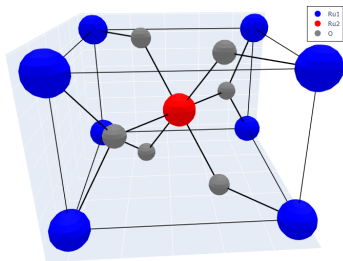
→ Wannierization: maximize the localization of $\tilde{\sigma}^2$!



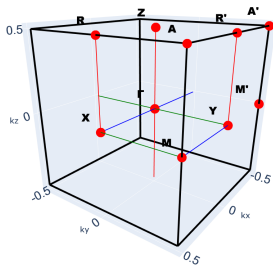
- DFT: Quantum Espresso
- Projection and wannierization: Wannier90
- Berry curvature calculation: PostWannier (utilities of Wannier90)

Results and Discussion

Optimized Structure of RuO_2 and Converged DFT Parameters



(a) Unit cell



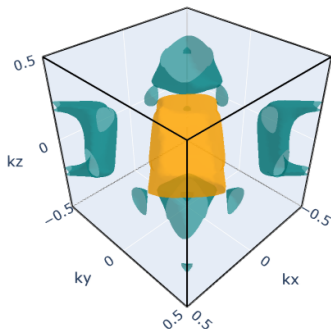
(b) Brillouin zone

Set of parameter using:

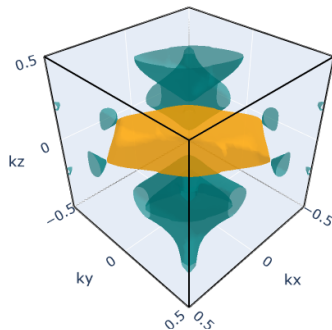
- Optimized lattice constants: $a = 4.5405 \text{ \AA}$, $c = 3.1323 \text{ \AA}$.
- energy wavefunction cutoff: $\text{ecutwfc} = 90 \text{ \AA}$.
- energy density cutoff : $\text{ecutrho} = 700 \text{ \AA}$.
- Hubbard U: 1.5 eV
- Scalar-relativistic calculations: no spin-orbit coupling.

Spin Polarized Fermi Surface

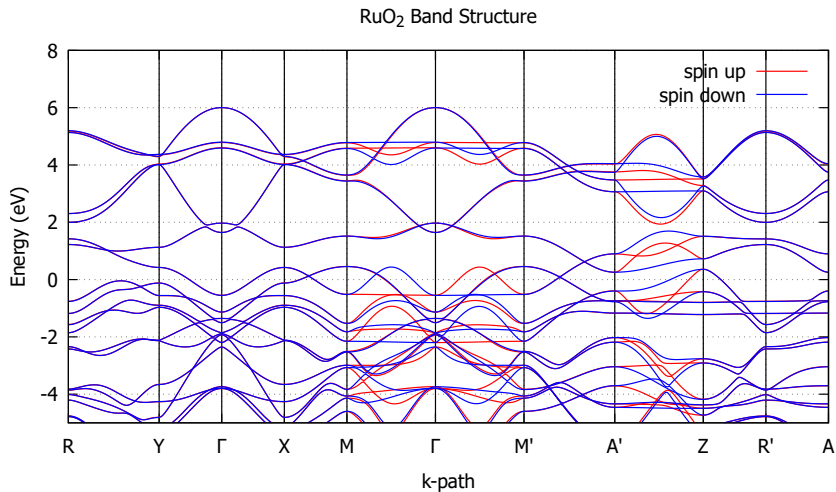
Spin Up



Spin Down

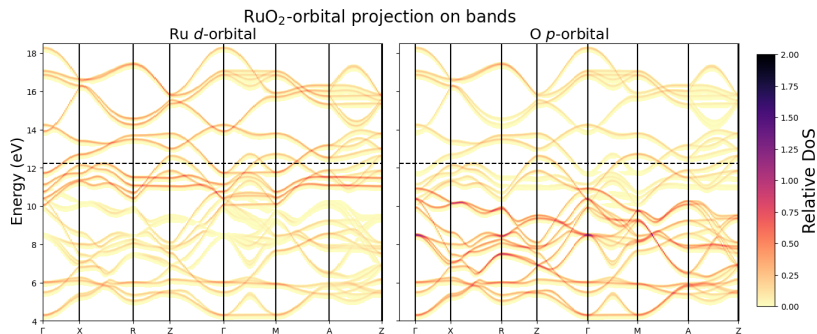


Spin-split Band Structure



→ Spin-split band structure without SOC included

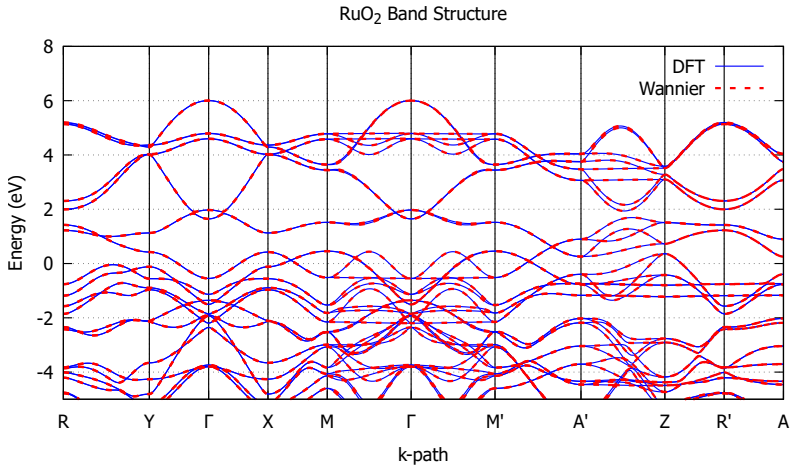
Selection of Orbitals For Wannierization



→ Using all Ru-d and O-p for Wannier projection.

Maximally Localized Wannier Functions:

After wannierization:

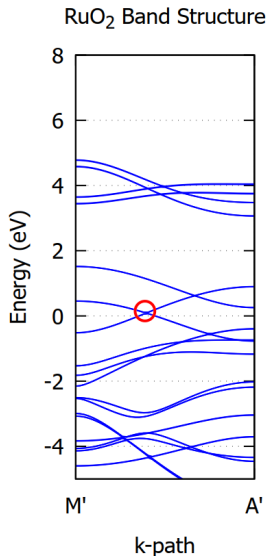


Match with DFT $\rightarrow$ Hamiltonian in Wannier basis!

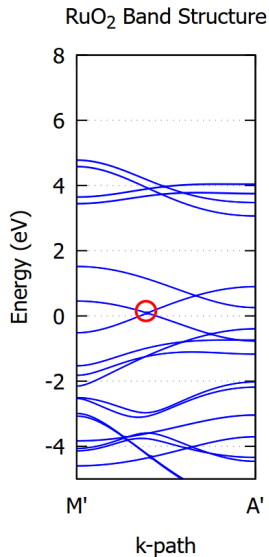
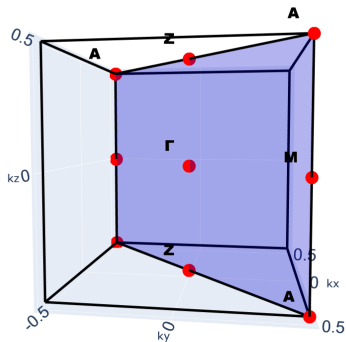
Berry Curvature Results

$$\Omega \propto \frac{1}{(E_n(\mathbf{k}) - E_m(\mathbf{k}))^2}$$

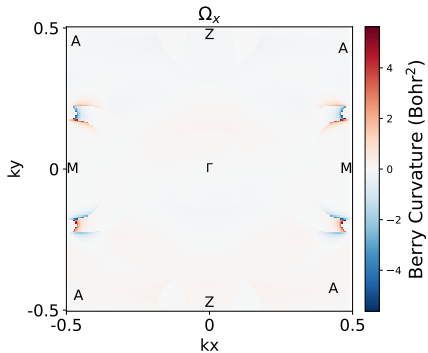
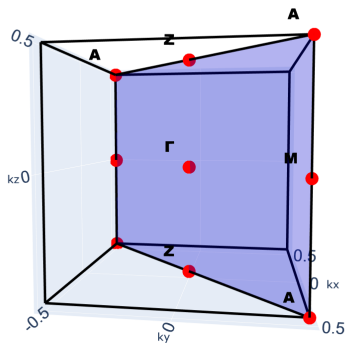
The closer the bands
The higher Berry curvature!



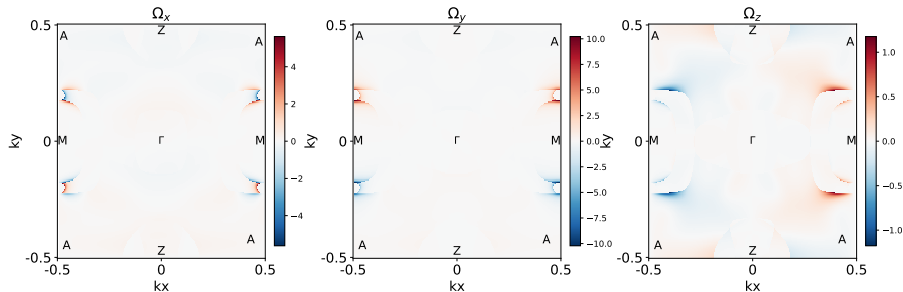
Berry Curvature Results



Berry Curvature Results



Berry Curvature Results



Anomalous Hall conductivity:

$$\sigma^{AHC} = \sum_{\mathbf{k}} \Omega(\mathbf{k})$$

Symmetry $\Omega(\mathbf{k}) \rightarrow$ No Conductivity!

How to Create High Conductivity?

Break the inversion symmetry

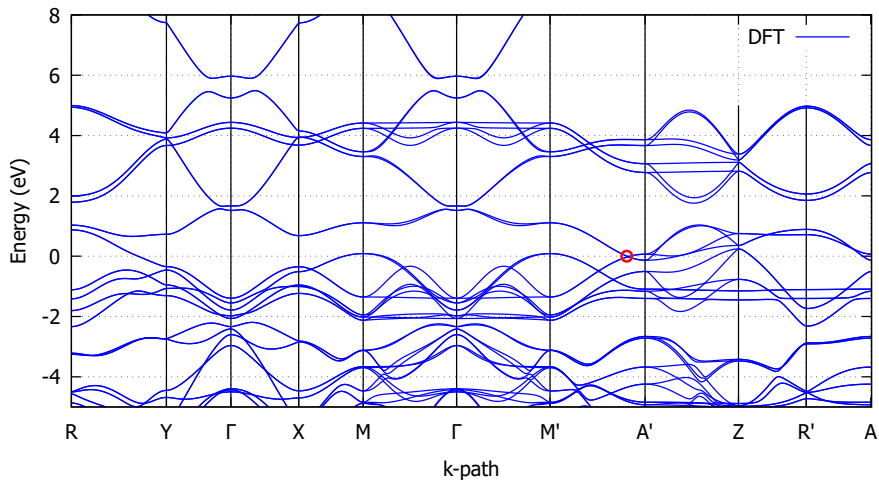
→ Applying the strain!

Can we increase Berry curvature by doping?

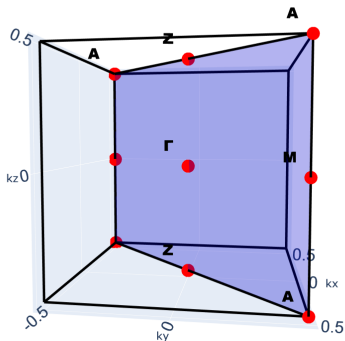
Electron doping + uniaxial strained systems!

Strain and Doped Band Structure

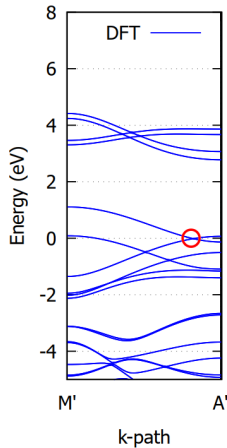
RuO₂ Strained Band Structure



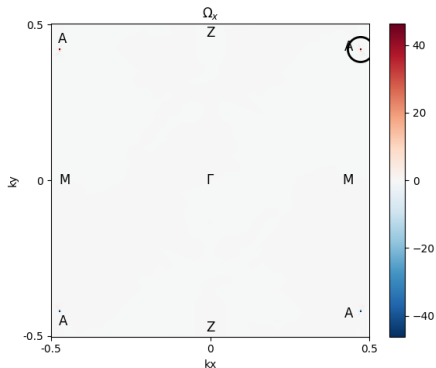
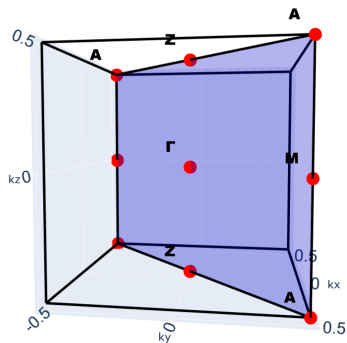
High Berry Curvature!



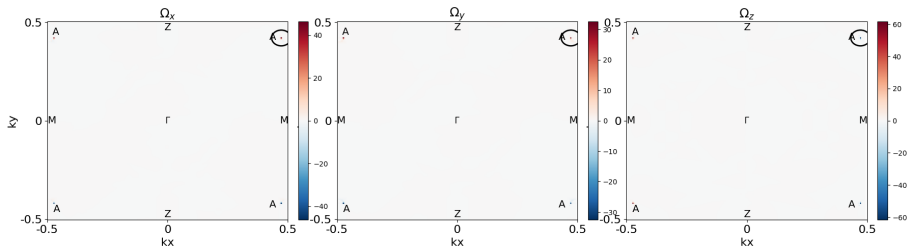
RuO₂ Strained Band Structure



High Berry Curvature!



Berry Curvature Under Strain and Doping



Capture the Dirac point $\rightarrow$ Higher Berry curvature!

Inversion symmetry breaking:

$$\rightarrow \sigma^{AHC} = \sum_{\mathbf{k}} \Omega(\mathbf{k}) \neq 0$$

Currently:

- Non zero result for low k-mesh
- Increasing the k-mesh to achieve the accurate result: Calculating

Summary

In this work:

- Confirmed the spin-polarized band structure without SOC.
- Calculated the Berry curvature.
- Modified the system to achieve higher Berry curvature.

Next steps:

- Anomalous Hall conductivity, Kubo optical response,...
- Include spin-orbit coupling
- k-space spin texture

Thank You!

- **Dr. Natasa Stojic** — my supervisor, for her continuous guidance.
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- **ICTP** people in general for the stimulating environment.